2503U2m2 math

1. The function f is defined by $f(x) = 6x - \frac{1}{8}$.
   What is the y-intercept of the graph of $y = f(x)$ in the xy-plane?
   A. $\left(0, -\frac{1}{8}\right)$
   B. $(0, -6)$
   C. $(0,6)$
   D. $(0,8)$

2. Which expression is equivalent to $(57y)^{\frac{1}{2}}$, where $y > 1$?
   A. $57\sqrt{y}$
   B. $\sqrt{57}y$
   C. $\sqrt{57y}$
   D. $\sqrt{(57y)^2}$

3. Luis has \$150 in an account. Each year he expects to have 1.1% more money in the account than he had in the previous year. Which of the following models best describes how Luis expects the money in the account to change over time?
   A. Decreasing exponential
   B. Decreasing linear
   C. Increasing exponential
   D. Increasing linear

4. The function m is defined by $m(x) = 5x + 3$, and the function p is defined by $p(x) = 3 - x$. What is the value of $2m(3) - p(3)$?

5. The number of bacteria in a liquid medium doubles every day. There are 48,000 bacteria in the liquid medium at the start of an observation. Which of the following represents the number of bacteria, y, in the liquid medium t days after the start of the observation?
   A. $y = \frac{1}{2}(48000)^t$
   B. $y = 2(48000)^t$
   C. $y = 48000\left(\frac{1}{2}\right)^t$
   D. $y = 48000(2)^t$

6. The solution to the given system of equations is (x, y). What is the value of $2y$?

$$y = \frac{x}{4} + 6$$
$$y = -\frac{x}{4} + 30$$

   A. 6
   B. 24
   C. 30
   D. 36

7. The function f and g are defined by the equations shown. Which of the following is equivalent to $f(x) - g(x)$?

$$f(x) = (2x - 3)(x + 8)$$
$$g(x) = 3(6x - 9)$$

   A. $2x^2 + 7x - 15$
   B. $2x^2 - 5x + 3$
   C. $2x^2 - x - 3$
   D. $2x^2 + 31x - 51$

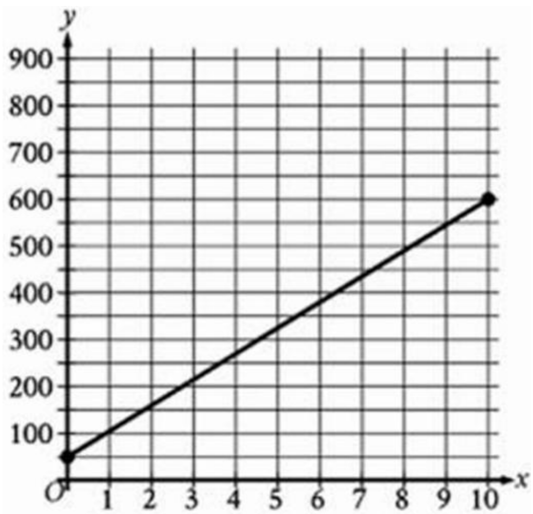
8. In right triangle ABC, angles A and B are acute, side AC has a length of 23.3, and $tanB = \frac{1}{4}$. What is the length of side BC, rounded to the nearest tenth?
   A. 4.8
   B. 5.8
   C. 93.2
   D. 542.9

9. What are all the solutions to the given equation?

$$-3|5x+7|+9=-12$$

A. 0
B. 0 and $-\frac{6}{5}$
C. 0 and $-\frac{14}{5}$
D. There is no solution.

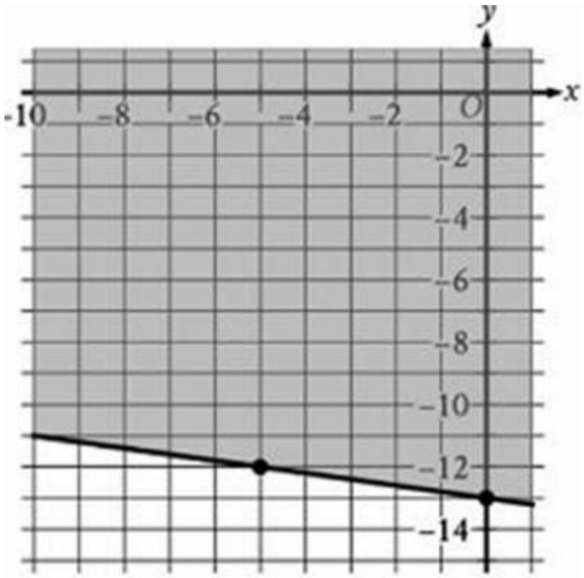
10. What is the slope of the line below?



11. The positive number a is $2{,}106\%$ of the sum of the positive numbers b and c, and b is 81% of c. What percent of b is a?

A. 4,706%
B. 2,600%
C. 38.12%
D. 21.87%

12. The shaded region shown represents the solutions to $rx + ty \geq -65$, where r and t are constants. What is the value of $r + t$?

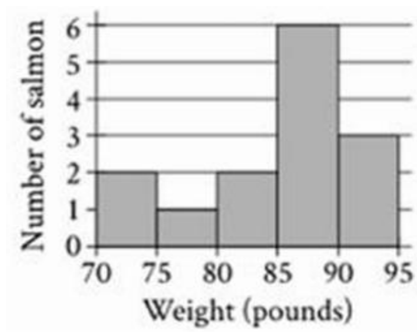


A. 6
B. 4
C. -4
D. -5

13. In the xy-plane, line k and line l are perpendicular and intersect at the point (7,5). If line k is defined by $y = mx + b$, where m and b are constants and $m > 1$, which of the following points lies on l?

A. $(8, 5 - \frac{1}{m})$
B. $(8, 5 + \frac{1}{m})$
C. $(8, 5 - m)$
D. $(8, 5 + m)$

14. The histogram summarizes a data set of weights, in pounds, of 14 salmon. If an additional weight of 32 pounds is added to the original data set to create a new data set of 15 weights, which of the following measures must be less for the new data set than for the original data set?



I. The median
II. The mean

A. I only
B. II only
C. I and II
D. Neither I nor II

15. For the function f, for every increase of 2 in the value of x, the value of $f(x)$ increases by a factor of c, where c is a constant. Which of the following equivalent forms of function f displays the value of c as the base or coefficient?

A. $f(x) = 42(2)^{6x}$

B. $f(x) = 42(8)^{2x}$

C. $f(x) = 42(64)^{x}$

D. $f(x) = 42(4096)^{\frac{x}{2}}$

16

To investigate the effect of magnesium supplementation on the sleep quality of adults, a researcher selected 190 adults at random from a community center to participate in a study. The researcher first measured the sleep efficiency of each participant. Each participant was then randomly assigned to take a magnesium supplement or a placebo each day for 6 weeks. At the end of the 6 weeks, the researcher measured the sleep efficiency of all participants again and found that the magnesium supplement caused statistically significant improved sleep quality for the adults at the community center. What feature of this study allowed the researcher to conclude that the magnesium supplement caused the improved sleep quality?

A) There were more than 100 participants in the study.

B) The participants were selected at random from the community center

C) The sleep efficiency of each participant was measured again at the end of 6 weeks.

D) Each participant was randomly assigned to take a magnesium supplement or a placebo.

17. A circle in the xy-plane has its center at $(-1,1)$. Line t is tangent to this circle at the point $(7,-6)$. Which of the following points also lies on line t?

A. $\left(0,\frac{8}{7}\right)$
B. $(6,9)$
C. $(14,2)$
D. $(15,1)$

18. The function f is defined by $f(x)=\frac{x^2+ax+b}{2x+c}$, where $a, b,$ and c are constants. The graph of the function f in the xy-plane does not intersect the line $x=4$. If $f(5)=f(6)=0$, what is the value of $a+b+c$?

19. If $4x+7=15$, what is the value of $8x-3$?

20. In the given equation, k is an integer constant. If the equation has two distinct real solutions, what is the greatest possible value of k?

$$kx^2 - 16x = 24x^2 - 8$$

21. Right rectangular prism X is similar to right rectangular prism Y. The surface area of right rectangular prism X is 56 square centimeters (cm^2), and the surface area of right rectangular prism Y is 896 cm^2. The volume of right rectangular prism Y is 1,536cm^3. What is the sum of the volumes, in cm^3, of right rectangular prism X and right rectangular prism Y?

22. In triangle ABC, the measure of angle B is 90° and BD is an altitude of the triangle. The length of AB is 24 and the length of AC is 13 greater than the length of AB. What is the value of $\frac{BC}{BD}$?

A. $\frac{13}{24}$

B. $\frac{24}{37}$

C. $\frac{37}{24}$

D. $\frac{24}{13}$